

Time Series Project — Phase 3

Task 1 — Data Exploration

Property	Value
Total number of clips	8732
Number of classes	10
Sampling rate (Hz)	22050
Min clip duration (s)	0.055
Max clip duration (s)	4.000
Mean clip duration (s)	3.608

Note: Class distribution is imbalanced — gun_shot (~370) and car_horn (~430) are under-represented vs. other classes (~1000).

Task 2 — Spectrogram Generation

Observation: STFT concentrates detail in low frequencies; Mel spreads detail uniformly. Transients show vertical bursts, sustained sounds show horizontal bands.

Task 3 — Transfer Learning Results (n_fft = 1024)

Spectrogram Type	Train Accuracy	Val Accuracy	Test Accuracy	Verdict
STFT	84.84%	72.79%	72.28%	
Mel	88.29%	74.39%	76.70%	Mel ✓

Comment: Mel outperforms STFT by ~4 percentage points on test accuracy.

Confusion matrix: see notebook Cell 30 (STFT and Mel test-set confusion matrices).

Task 4 — Window Size Experiment

Window Size (n_fft)	STFT Train	STFT Test	Mel Train	Mel Test
256	85.00%	74.07%	81.75%	73.95%
512	83.57%	72.64%	82.82%	71.92%
1024 (default)	86.35%	75.03%	85.87%	77.54%
2048	80.18%	68.94%	88.26%	76.70%

Q1 — Shape effect: Smaller n_fft → more time frames, fewer frequency bins. Larger n_fft → the opposite (time-frequency uncertainty principle).

Q2 — Best window per type: STFT → n_fft = 1024 (75.03%). Mel → n_fft = 1024 (77.54%).

Q3 — Consistent trend: Yes. Both peak at n_fft = 1024, and Mel matches or beats STFT at every window size.

Project Result

In Phase 3, I implemented a transfer learning pipeline using a frozen ImageNet-pretrained EfficientNet-B0 with only the final 10-class head trained on UrbanSound8K spectrograms. I compared STFT and Mel representations across four window sizes (256, 512, 1024, 2048). Mel matched or outperformed STFT at every window size tested, and both spectrogram types peaked at n_fft = 1024 — STFT at 75.03% and Mel at 77.54%. The overall best configuration was Mel at n_fft = 1024, confirming the perceptual advantage of mel-scaled features for pretrained image backbones: spreading useful information uniformly across the image transfers better than concentrating it in a thin low-frequency band.